

Haibin Guan

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RESEARCH INTERESTS

Metabolomics • Environmental Exposomics • Multi-omics Integration • High-dimensional Mediation Analysis • Statistical Methods for Omics Data

WORK EXPERIENCE

Icahn School of Medicine at Mount Sinai New York, United States
Data Analyst I 5/2022 – 5/2024
Data Analyst II 5/2024 – Present

- Conduct metabolomics and exposomics analyses including data preprocessing, normalization, and statistical modeling on high-dimensional omics datasets.
- Collaborate with biostatistics and environmental health research teams on method development and advanced statistical analyses for epidemiologic studies.

INFOTECHSoft Miami, FL, United States
Biomedical Data Analyst Intern 1/2018 – 7/2018

- Contributed to the development of a metabolomics data integration system for multi-platform biological data.
- Collaborated with bioinformatics researchers and software engineers to support large-scale microbiome data integration and analysis.

EDUCATION

Florida International University Miami, FL, United States
MS in Data Science 1/2017 – 8/2018
Transferred to MS in Computer Science 8/2018 – 5/2019
Wenzhou Medical University Wenzhou, Zhejiang, China
Bachelor in Preventive Medicine 9/2011 – 7/2016

SELECTED RESEARCH PROJECTS

Multi-omics Integration Analysis of Environmental Exposures and Cancer Risk 2025 – Present
Icahn School of Medicine at Mount Sinai

- Evaluating multi-omics data integration approaches to identify latent molecular components linking metabolomic and proteomic profiles with thyroid cancer risk and PFOS exposure.
- Evaluating stability and reproducibility of latent molecular signatures using resampling-based approaches.
- Manuscript in preparation.

Identification of Critical Windows of Susceptibility Using Time-varying Exposure Models 2025 – Present
Icahn School of Medicine at Mount Sinai

- Applying statistical models for time-varying exposures/mixtures, including BDLIM, TDLMM or other tree-based distributed lag models (dlmtree), to identify critical windows of susceptibility.
- Investigating associations between longitudinal environmental exposures and maternal health outcomes while characterizing sensitive developmental periods.
- Manuscript in preparation.

Maternal PFAS Mixture Exposure and SARS-CoV-2 IgG: Metabolomic Mediation Analysis 11/2023 – 12/2024
Icahn School of Medicine at Mount Sinai

- Conducted high-dimensional mediation analysis to examine whether maternal metabolomic profiles mediate the association between PFAS exposure and SARS-CoV-2 IgG levels during pregnancy.
- Constructed PFAS mixture indices using weighted quantile sum (WQS) regression and evaluated metabolite mixture effects within mediation models.

Comparison of Metabolomic Profiles Across DBS, DBM, and Plasma Matrices

04/2023 – 05/2024

Icahn School of Medicine at Mount Sinai

- Compared untargeted metabolomic measurements across dried blood spots (DBS), dried blood microsamplers (DBM), and plasma samples.
- Evaluated cross-matrix consistency and demonstrated moderate agreement, with DBS and DBM showing comparable metabolomic profiles.

Metabolomics Data Processing and Batch Correction Pipeline

02/2023 – 01/2024

Icahn School of Medicine at Mount Sinai

- Developed automated preprocessing workflows for untargeted metabolomics data using R-based tools (XCMS, IPO) for peak detection, retention time alignment, and feature grouping.
- Evaluated batch correction and missing-data imputation strategies and generated automated quality-control reports for LC/MS and GC/MS datasets.

PFAS Exposure and Thyroid Cancer Risk

10/2022 – 03/2023

Icahn School of Medicine at Mount Sinai

- Investigated associations between PFAS exposures and thyroid cancer risk using plasma metabolomics data from case-control pairs in the BioMe biobank.
- Applied regression-based epidemiologic models with covariate adjustment and inverse probability weighting to address potential selection bias.

DATA SCIENCE COMPETITIONS

TReNDS Neuroimaging Kaggle Competition

4/2020 – 6/2020

Participant

Kaggle

- Built machine learning models to predict brain age and cognitive assessment scores using functional connectivity and structural MRI features.
- Applied PCA-based dimensionality reduction and trained regression models including SVR, KNN, and ElasticNet.
- Achieved a Silver Medal ranking (27 / 1047 teams).

Multiple Myeloma DREAM Challenge

7/2017 – 12/2017

Participant

Florida International University

- Developed predictive models for gene expression datasets using feature selection and ensemble machine learning approaches.
- Applied SMOTE sampling, differential expression analysis (limma), and multiple classifiers including SVM, Random Forest, and Neural Networks.
- Achieved first place in the second round of the competition among 40 teams.

TECHNICAL SKILLS

Programming: R, Python, MATLAB, Linux, LaTeX, XML/HTML, CSS

Statistical Methods: Regression Modeling, Cox Models, High-dimensional Mediation Analysis, Weighted Quantile Sum Regression

Metabolomics Tools: XCMS, GNPS, MetaboAnalyst

Microb Tools: QIIME2, Kraken, PICRUSt

Deep Learning Frameworks: TensorFlow, PyTorch

Cloud Computing: AWS

PUBLICATIONS

- **Guan H**, van Gerwen M, Kim-Schultz S, Colicino E, Dolios G, Petrick LM. Stability-driven multi-omics integration for reproducible latent structure. *bioRxiv*. 2026.06.23.734064.
- Midya V, Maroli AS, Picker M, Dolios G, Valvi D, Nguyen I, Kim T, Rendon A, Chen R, Weinstein K, **Guan H**, Joseph G, Eggers J, Torres-Olascoaga LA, Ravisekar R, Srinath B, Ranchadiya R, Andra SS, Achaintre D, Gennings C, Téllez-Rojo MM, Wright RO, Arora M, Rosa MJ, Niedzwiecki M, Torres J, Alcala CS, Lane JM, Eggers S, Colombel JF, Peter I, Petrick LM, Agrawal M. Per- and polyfluoroalkyl substances (PFAS) in early life is associated with childhood intestinal inflammation: analyses of three birth cohorts. *Clin Gastroenterol Hepatol*. 2026.

- Petrick LM, Dassanayake PS, Gennings C, **Guan H**, Akintunde B, Martinez M, Regmi P, Hertz-Picciotto I, Arora M, Bennett DH. Fetal exposure to persistent organic pollutants and childhood autism risk. *Environ Res.* 2026;125179.
- Svensson K, India-Aldana S, Saddiki H, Petrick LM, **Guan H**, Hernández-Chávez C, Téllez-Rojo MM, Wright RO, Wright RJ, Colicino E. Pregnancy Stress Exposures and Postpartum Serum Metabolomic Profiles in Mothers. *Metabolites.* 2026;16(5):312.
- Joseph G, Renard-Dausset E, Invernizzi A, Chen J, **Guan H**, Wright RJ, Wright R, Rosa MJ, Horton M, Téllez-Rojo MM, Torres Calapiz M, Colicino E, Vrijheid M, Foraster M, Bustamante Pineda M, Rivas I, Llurba Olivé E, Maitre L, Petrick LM. Association between per- and polyfluoroalkyl exposures and postpartum psychological symptoms: Evidence from two population-based pregnancy cohorts. *Environ Res.* 2026;305:124923.
- Metayer C, Gennings C, Morimoto LM, Mistry S, McGlynn KA, Dolios G, **Guan H**, Gonzaga CR, Andra SS, Petrick LM. Phthalates measured at birth and risk of testicular cancer in adolescents and young adults. *JNCI Cancer Spectrum,* 2026.
- Svensson K, Sanchez-Guerra M, Midya V, **Guan H**, Niedzwiecki M, Dolios G, Osorio-Yáñez C, Del Razo LM, Petrick LM, et al. Postpartum metabolomics profile predicts women's subclinical atherosclerosis. *Scientific Reports,* 2025.
- **Guan H**, Chen J, Kaur K, Amreen B, Lesseur C, Dolios G, Andra SS, Narasimhan S, Pulivarthi D, Midya V, De Witte LD, Bergink V, Rommel AS, Petrick LM. High-dimensional mediation analysis to elucidate the role of metabolites in the association between PFAS exposure and reduced SARS-CoV-2 IgG in pregnancy. *Sci Total Environ.* 2025.
- Morimoto LM, Metayer C, Dolios G, Wiemels JL, Ma X, **Guan H**, Petrick LM. PFAS exposure during pregnancy and risk of childhood leukemia. *Environmental Research,* 2025.
- Joseph G, van Gerwen M, Kim-Schulze S, Chen Z, Nadkarni GN, **Guan H**, Petrick LM. Inflammatory proteins and thyroid cancer risk. *International Journal of Cancer,* 2025.
- Petrick LM, Niedzwiecki MM, Dolios G, **Guan H**, Tu P, Wright RO, Wright RJ. Effects of storage conditions on metabolomics profiles. *Sci Total Environ,* 2024.
- Petrick LM, **Guan H**, Page GP, Dolios G, Niedzwiecki MM, Wright RO, Wright RJ. Comparison of maternal blood metabolomics across DBS, DBM, and plasma. *Environment International,* 2024.
- van Gerwen M, Colicino E, **Guan H**, Dolios G, Nadkarni GN, Vermeulen RCH, Wolff M, Arora M, Genden EM, Petrick LM. PFAS exposure and thyroid cancer risk. *SSRN.*
- Soto P, Ilmer I, **Guan H**, Li J. Lightweight projective derivative codes for compressed asynchronous gradient descent. *ICML,* 2022.
- Soto P, **Guan H**, Li J. Locally Random P-adic Alloy Codes with Channel Coding Theorems for Distributed Coded Tensors. *arXiv.*
- Ruiz-Perez D, **Guan H**, Madhivanan P. So you think you can PLS-DA?. *BMC Bioinformatics,* 2020.
- Zhang T, **Guan H**, Li F, He F. Modeling influenza-like illness prediction based on Elman neural network. *Preventive Medicine,* 2019.

PRESENTATIONS

- High-dimensional mediation analysis to elucidate the role of metabolites in the inverse association between PFAS exposure and SARS-CoV-2 IgG in pregnancy. MANA 2024 Plenary Presentation, Tampa, FL.

SOFTWARE AND APPLICATIONS

- Worklists Generator: R Shiny application designed to generate Orbitrap instrument worklists from SampleID files. https://haibin-guan.shinyapps.io/worklists_generator/
- BatCheck: R Shiny application for visualization and assessment of inter- and intra-batch effects in mass spectrometry data. <https://haibin-guan.shinyapps.io/BatCheck/>

PROFESSIONAL DEVELOPMENT

Multi-Omics Boot Camp	New York, NY, USA
Participant, Columbia University Multi-Omics Initiative	07/2026
MANA WomiX Mentorship Program	USA
Participant	4/2024 – 2/2025
Oxford Machine Learning Summer School	Oxford, United Kingdom
Participant	7/2023